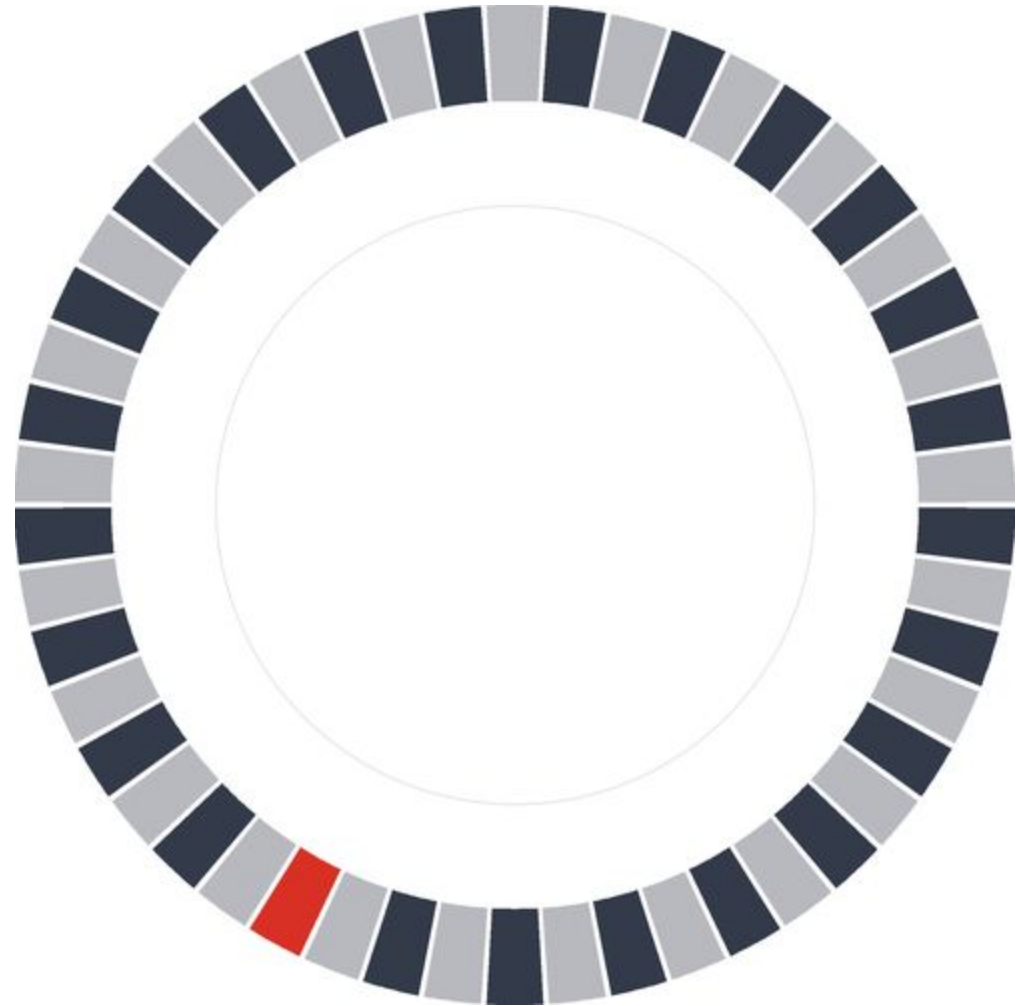


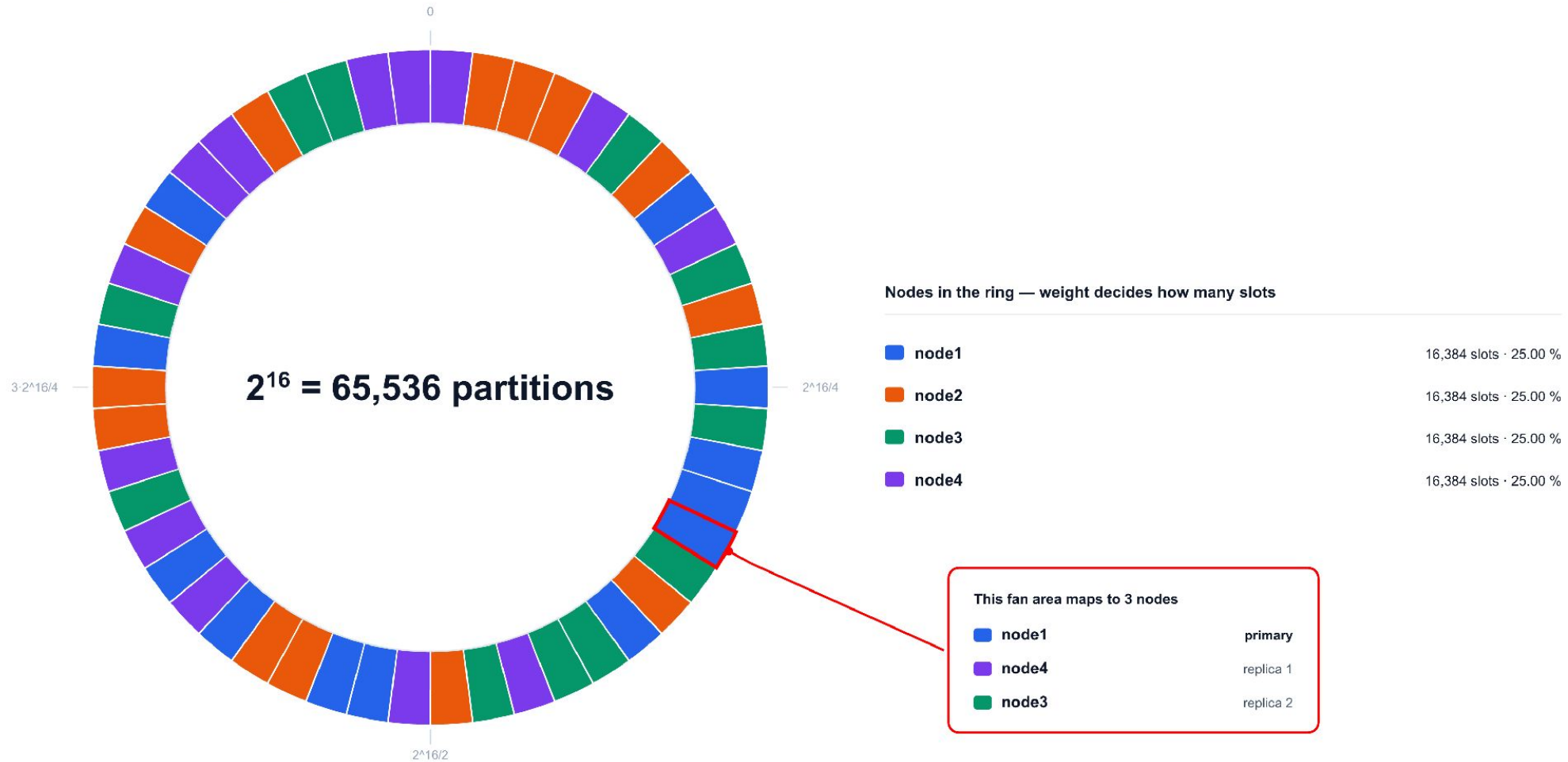
OpenStack Swift

Consistent ring bug fix & poster

TEAM	Highly Available Object Storage Development Abdoul Abdou · Bowen Tan · Yihan Wang · Jiabao Chen
MENTOR	Andressa Cabistani — Red Hat
TARGET	Launchpad #1693149 — “RingBuilder.save isn’t atomic”

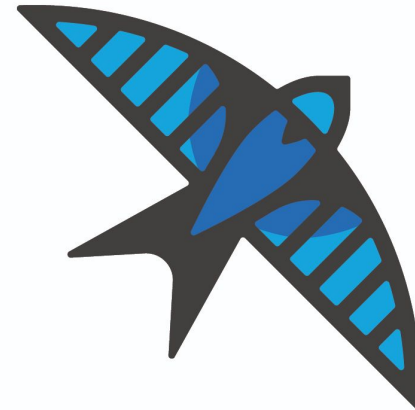


Consistent hashing ring



High level idea of consistent hashing

- Account John_Smith wants to store EC528_Cloud_Computing/Demo1.pptx into Swift.
- `partition_key = John_Smith_EC528_Cloud_Computing/Demo1.pptx`
- `hash(partition_key) % 2^16 = 52828`
- `ring[52828] = {node1, node4, node3}`

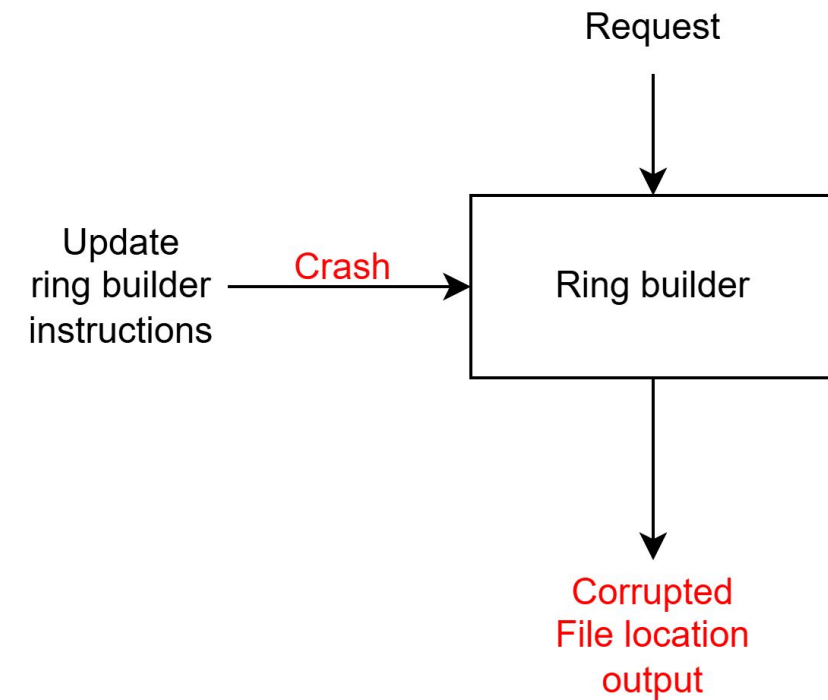


SWIFT

an OpenStack Community Project

RingBuilder.save() is not atomic

- RingBuilder.save() rewrites that file in place.
- A crash or a power loss mid-write leaves the file half-written.
- Following lookup would fail due to a corrupted ring.



How we work, and how we know it worked

PLAN

Ring bug — reproduce the bug on local environment. Modify the ring in a temp file. Swap the name of RingBuilder and RingBuilderTemp

Poster — comparing swift's consistent hashing ring and RDBMS(postgreSQL's) share nothing technology on scalability and fault tolerance.

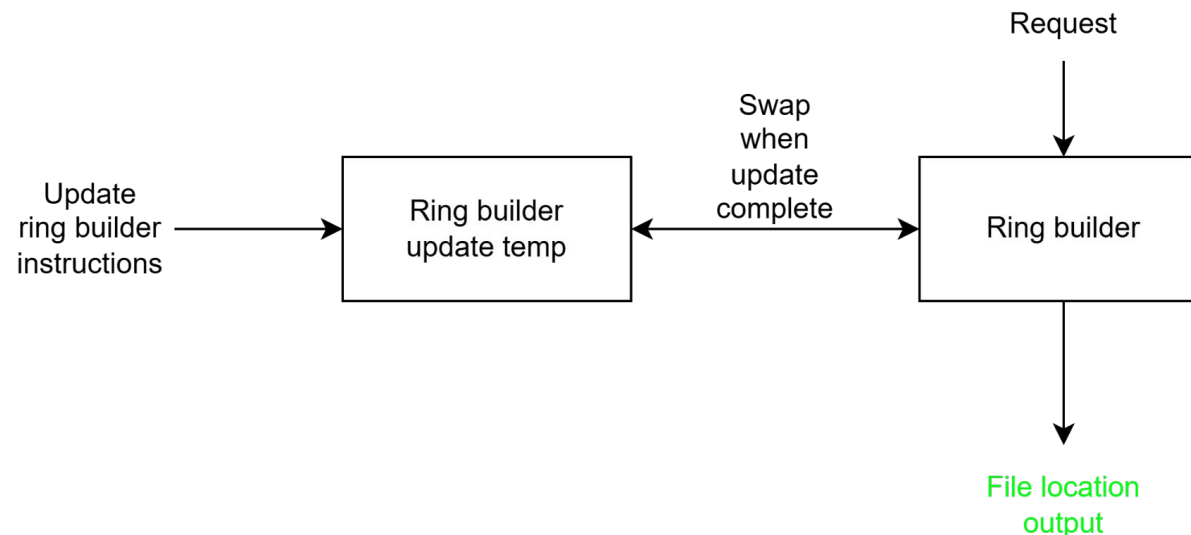
Forum follow-up — resolve any follow up questions on the bug forum.

METRICS

Ring bug — Run the same kill-mid-update against the patched builder; the composite ring builder is byte-identical to its pre-crash contents and ring position lookups still succeed.

Poster — Mentor's PR approval on the poster.

Forum follow-up — The Launchpad bug report for the composite ring builder is marked Fix Released.



Why this is hard

01 ·

The ring builder is the core of traffic control for the entire system.

If it fails, the cluster loses the map that decides where every byte lives — permanent data loss.

02 ·

A corrupted ring weakens Swift's consistency even more.

Swift is an eventually consistent system, that relies on high availability, a bad write's effects can be seen by many users easily.

Thanks

